

B.C.A. 3rd Semester Full and Reappear

Examination, November-2023

INTRODUCTION TO OPERATING SYSTEM

Paper-BCA-201

Time allowed : 3 hours] [Maximum marks : 80

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Attempt five questions in all, selecting one question from each unit. Question No. 1 is compulsory.

1. (a) Real Time Operating System 4×4=16
- (b) Define Shortest Job First. 15
- (c) Explain Thrashing and how to overcome.
- (d) Define Disk Structure.

Unit-I

2. (a) Define operating system with its components. 8
- (b) What do you mean by operation on Process ? 8
3. Explain Inter-Process communication in detail. 16

Unit-II

4. (a) Define Deadlocks Preventions with example. 8
- (b) Explain Deadlocks Detection and Recovery. 8
5. Explain Scheduling. Also explain various scheduling algorithms. 16

Unit-III

6. (a) Write various contiguous memory allocation techniques. 8
- (b) Explain Demand Paging. 8
7. Differentiate : 15
- (a) Internal and External fragmentation. 8
- (b) Logical and Physical Address. 8

Unit-IV

8. Explain Disk Scheduling Algorithms in detail. 16
9. What do you understand by Free Space Management? 16